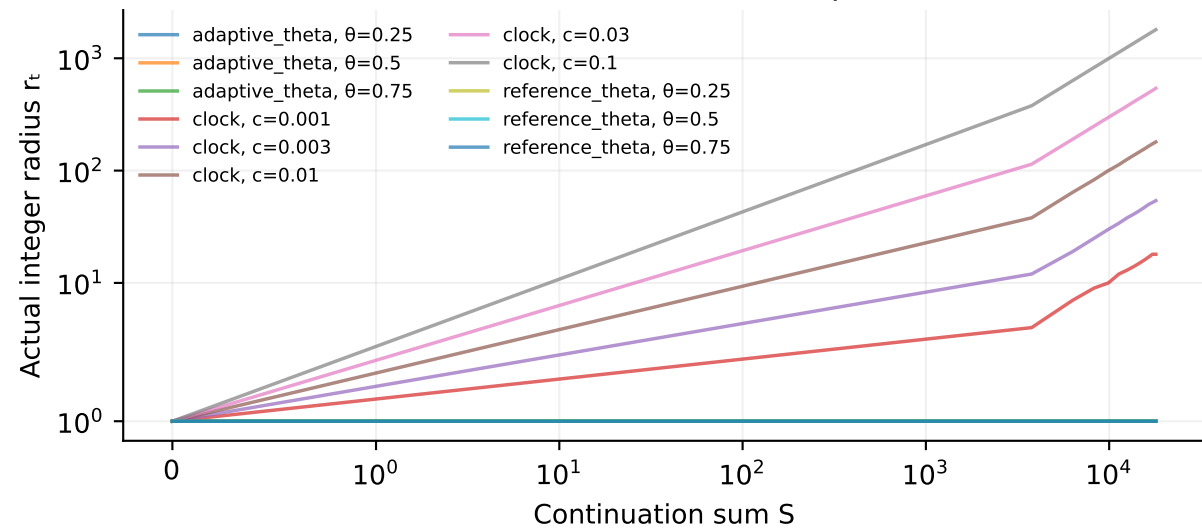
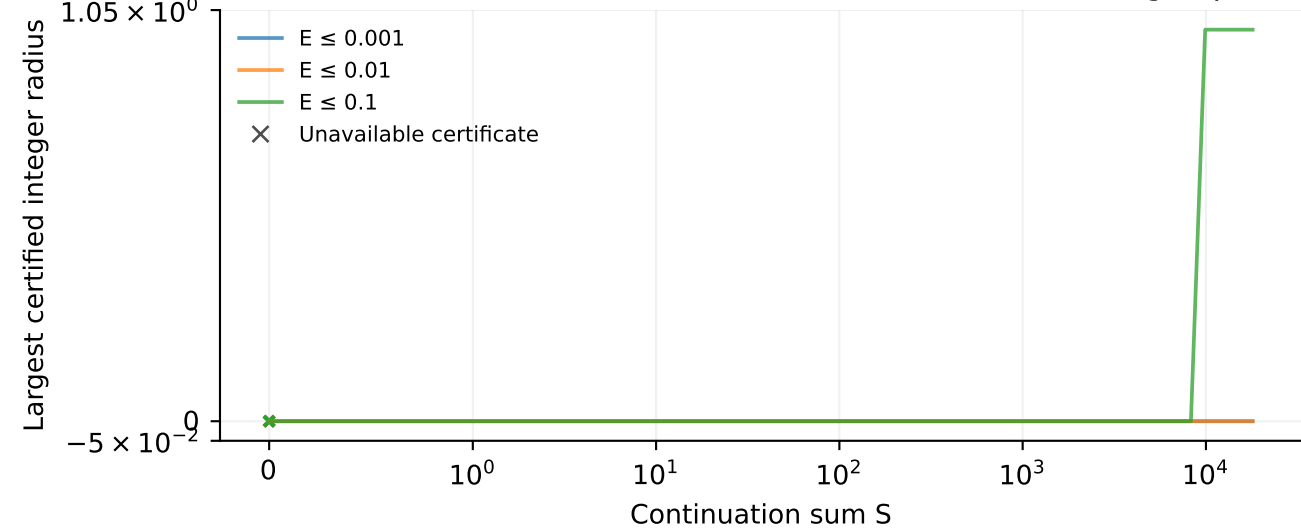


Measured radius trajectories and certified generalization range

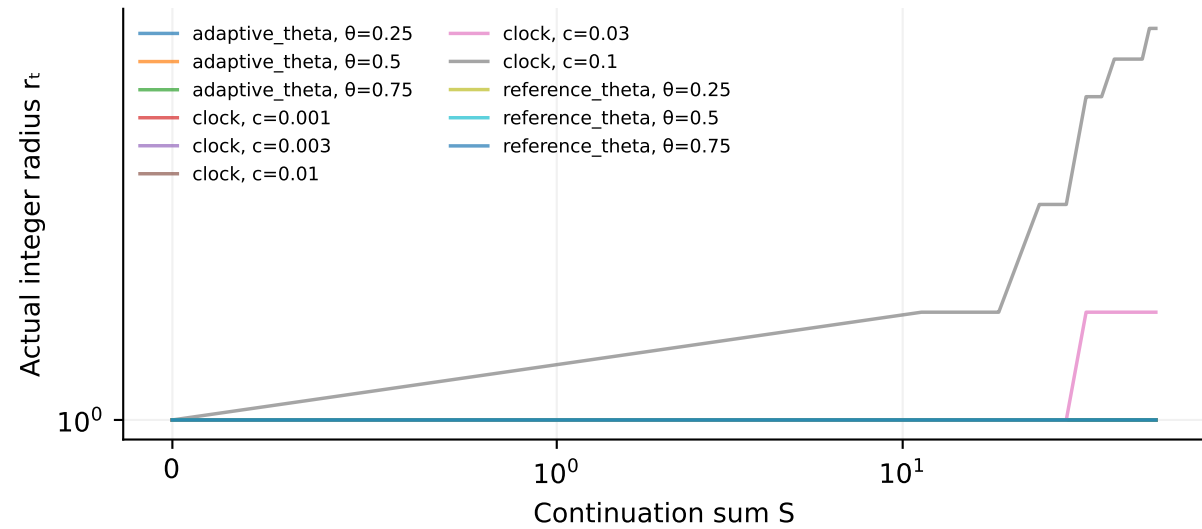
Learned retrieval · SGD — actual probe radius



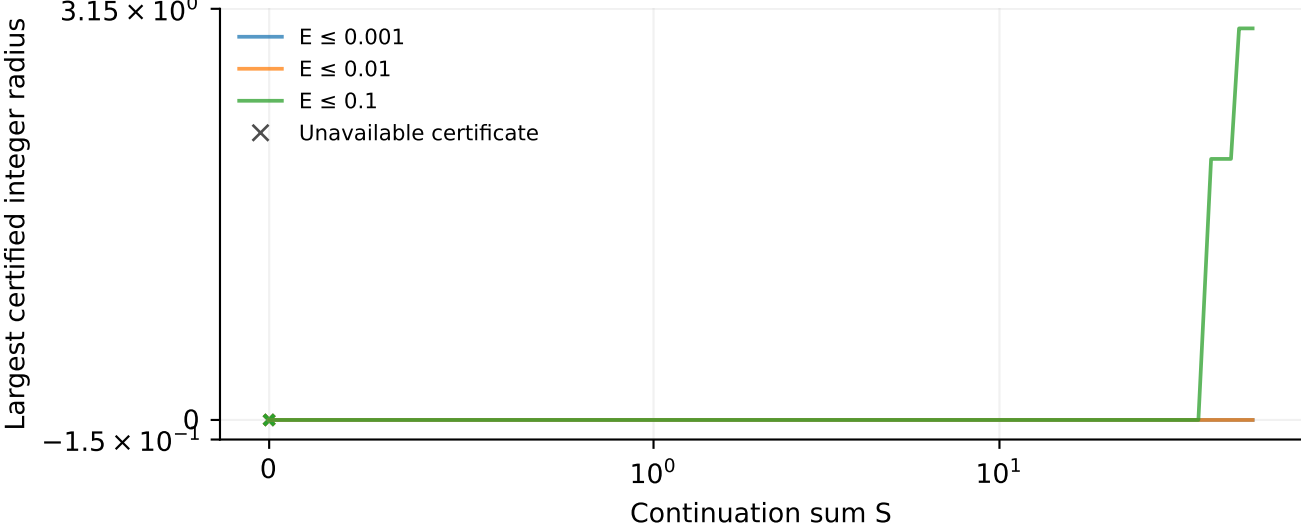
Certified radius from uniform bound; no evaluation-lag cap



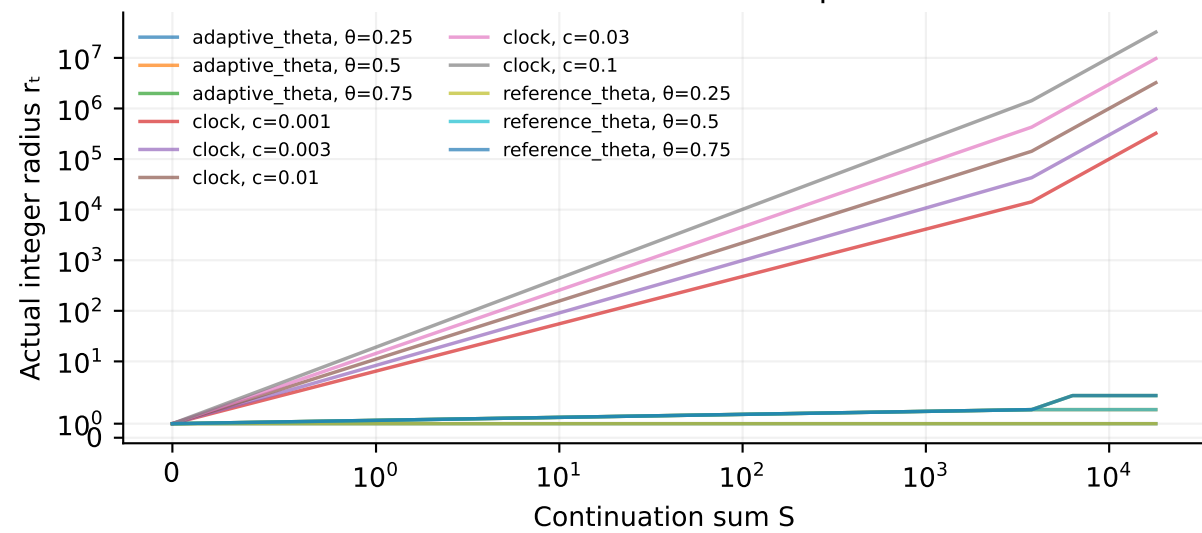
Learned retrieval · ADAM — actual probe radius



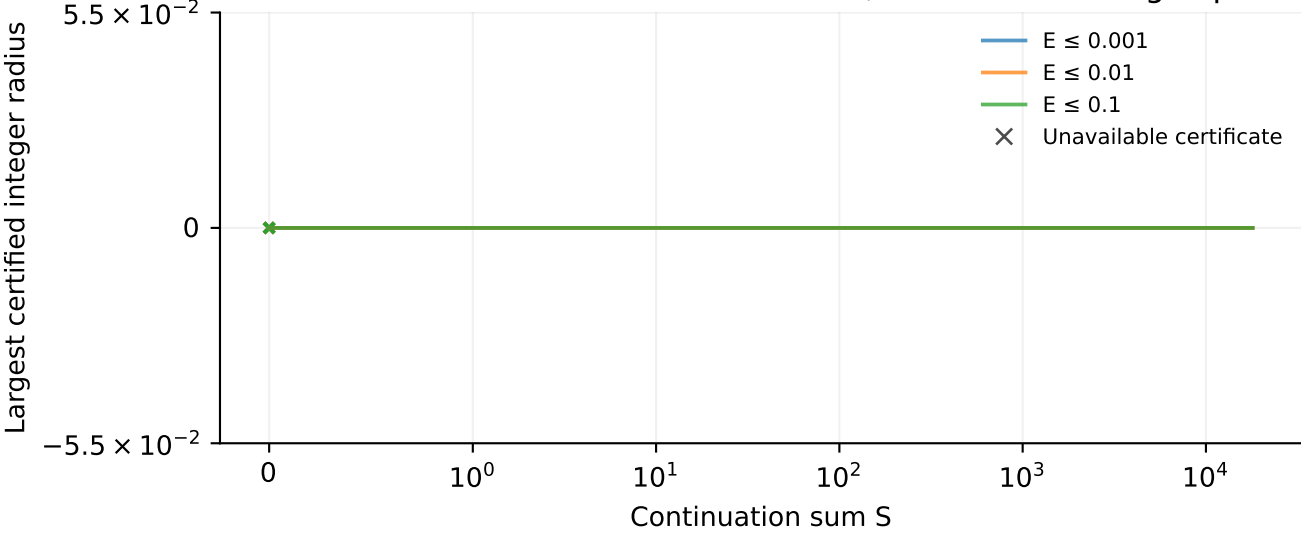
Certified radius from uniform bound; no evaluation-lag cap



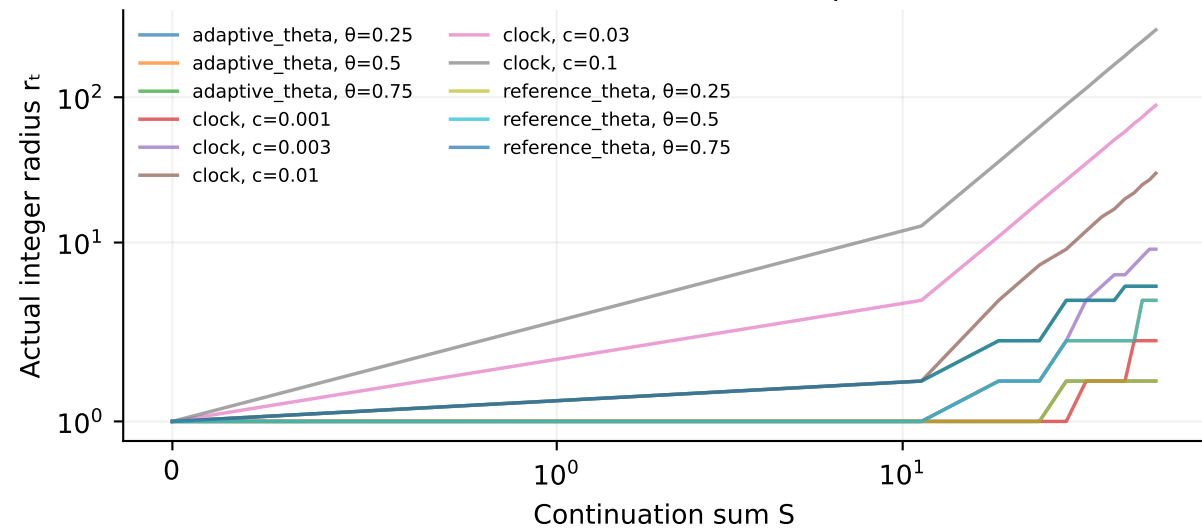
Frozen retrieval · SGD — actual probe radius



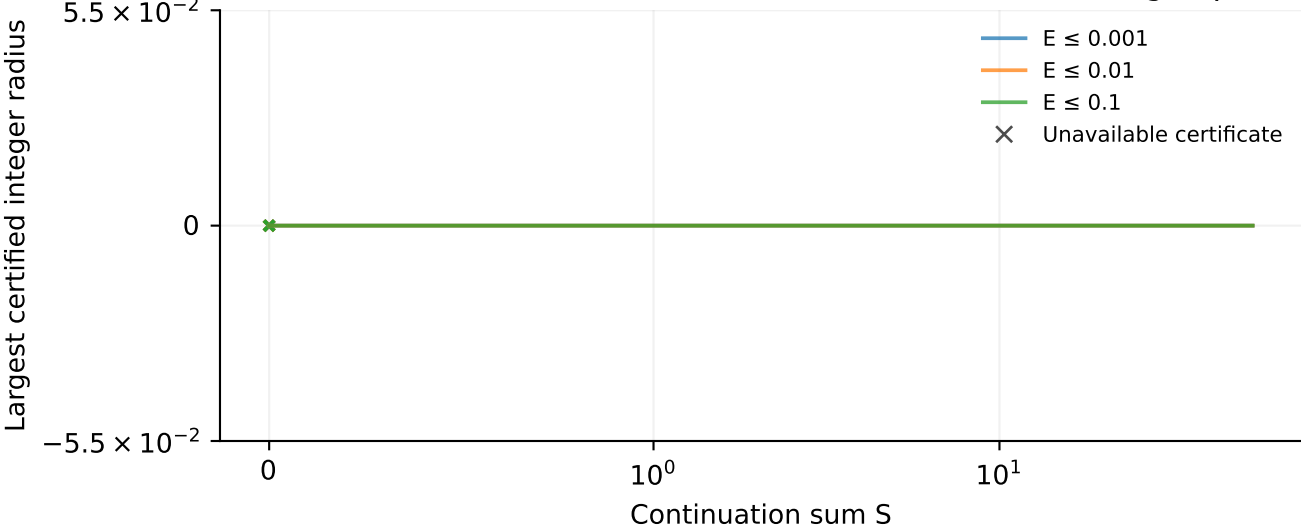
Certified radius from uniform bound; no evaluation-lag cap



Frozen retrieval · ADAM — actual probe radius



Certified radius from uniform bound; no evaluation-lag cap



Clock-based radii are chosen probes; adaptive radii use measured gaps. Observed growth over a finite window does not establish $r_t \rightarrow \infty$. Radius 0 certifies no positive radius.